# Compare and round parachute measures

A Coffee Filter Parachute

Name _____ Date _____

Write one answer in each blank. For comparisons, also put >, <, or = in the circle.

**1.** Across several drops, a crumpled filter floats for 28 counted seconds in all and a flat filter for 42. Put >, <, or = in the circle. What is the greater total?

28 ○ 42

Show your thinking.

**2.** One set of parachute strings is 76 centimeters long and another is 67 centimeters. Put >, <, or = in the circle. What is the greater length?

76 ○ 67

Show your thinking.

**3.** Big and Little each record 48 test drops. Put >, <, or = in the circle. What is the shared number when the totals are equal?

48 ○ 48

Show your thinking.

# Compare and round parachute measures

A Coffee Filter Parachute

Name _____ Date _____

Write one answer in each blank. For comparisons, also put >, <, or = in the circle.

**4.** The supply box holds 23 paperclips. What is 23 rounded to the nearest ten?

23 → nearest ten: ___

Show your thinking.

**5.** A piece of string is 57 centimeters long. What is 57 rounded to the nearest ten?

57 → nearest ten: ___

Show your thinking.

**6.** A safe standing-height drop is measured as 95 centimeters. What is 95 rounded to the nearest ten?

95 → nearest ten: ___

Show your thinking.

## A pull and a push

Gravity pulls the parachute down. Moving air pushes back with air resistance, or drag, slowing the fall without stopping gravity.





## Wide meets air

The flat filter and crumpled filter weigh the same. The wide shape meets more air, makes more drag, and falls more slowly.





## Change the weight

A second paperclip adds weight, so gravity pulls down harder while the canopy stays the same size. In air, it usually reaches the floor sooner.





## A fair test

Change one thing at a time, keep the others equal, and measure several drops. Repeating helps a result become more trustworthy.





## A steadier fall

A tiny hole at the canopy top lets some air through. The flow can become more even, so the parachute wobbles less.





## The same big idea

Human and space-lander parachutes use wide canopies too. Gravity pulls down while drag pushes against motion and slows the descent.



## Make a little book



**1** Fold it into 8 boxes. Open it flat.



**2** Fold short end to short end. Cut the thick line. Open it.



**3** Fold the long way, pictures out. Push the ends in.




**4** Fold into a book. Page 1 goes on the front!





WONDER DAILY

# A Coffee Filter Parachute

# Write a fair-test note

A Coffee Filter Parachute · Curiosity day

Name _____ Date _____

Choose one clue. Say your idea first, then draw or write it.



Wide catches more air



One change at a time



A steadier path

Draw one parachute test and add a small arrow to the part you changed.

gravity · air · air resistance · drag · canopy · surface area · weight · fair test · predict · measure

The wide filter fell more slowly because … dictate how it meets the air

To make the test fair, I would … dictate what stays the same

The tiny top hole may … dictate how the fall changes

Does your sentence name one change and explain what you would predict or measure?

# A Coffee Filter Parachute

## 01 What slows the parachute?



A wide canopy meets lots of air as gravity pulls the parachute down.

**Look closer.** Find the wide canopy above the hanging paperclip and notice the open space it covers.

**A little explanation.** Gravity pulls the parachute down. As it moves, tiny moving pieces of air push back. That air resistance, also called drag, slows the fall; it does not turn gravity off.

**Try this.** An adult stays beside you while you attach equal strings or yarn and a paperclip or small toy to a coffee filter. Ask that adult to use the scissors. Drop from standing height, or let the adult hold it higher. Never climb on chairs, stair railings, windows, or balconies.

## 02 Why does the flat filter fall more slowly?



The flat, wide filter has more surface area facing the air.

**Look closer.** Compare the open, wide filter with the same kind of filter crumpled into a small shape.

**A little explanation.** The wide filter has more surface area facing the moving air. It catches more air and makes more drag, so it falls more slowly than the crumpled filter even though both filters weigh the same.

**Try this.** Drop one filter flat and the same filter crumpled. Count the fall time or use a timer, then open the filter and repeat.

## 03 What changes when the load is heavier?



A second paperclip adds weight, so gravity pulls down harder while the same canopy provides drag.

**Look closer.** Look for one paperclip under one canopy and two paperclips under the matching canopy.

**A little explanation.** With the same canopy, a second paperclip adds weight, so gravity pulls down harder while the canopy area stays the same. In this test, that parachute usually descends faster. Without air, the extra paperclip would not make it fall faster.

**Try this.** Keep the filter and strings the same. Compare one paperclip with two paperclips over several drops.

## 04 How can a tiny hole make the fall steadier?



A tiny top hole can let air through and reduce side-to-side wobble.

**Look closer.** Find the small centered hole and watch the canopy stay open with less side-to-side wobble.

**A little explanation.** A small hole at the canopy top lets some air flow through. That can keep air from spilling unevenly around the edges, so the parachute may wobble less.

**Try this.** Predict what a tiny centered hole will change. Test the uncut canopy first, then ask an adult to make the hole and measure several more drops.